

① Shankar 17.2.2

$$\hat{H} = \hat{H}_0 + \hat{H}' \quad \hat{H}_0 = -\gamma B_0 \hat{S}_z \quad \hat{H}' = -\gamma B \hat{S}_x$$

Unperturbed states are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in $\hat{S}_z$ basis,
energies $-\frac{\hbar\gamma B_0}{2}$ and $+\frac{\hbar\gamma B_0}{2}$. $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \leftrightarrow |+\rangle$ ground state.

call states $|0^{(0)}\rangle$ and $|1^{(0)}\rangle$.

$$E_0^{(1)} = \langle 0^{(0)} | \hat{H}' | 0^{(0)} \rangle = -\gamma B \langle +_z | \hat{S}_x | +_z \rangle = 0$$

$$E_0^{(2)} = \sum_{i \neq 0} \frac{|\langle i^{(0)} | \hat{H}' | 0^{(0)} \rangle|^2}{(E_0^{(0)} - E_i^{(0)})}$$

$$= (\gamma B)^2 \frac{|\langle -_z | \hat{S}_x | +_z \rangle|^2}{-\hbar\gamma B_0} = \frac{-\hbar\gamma^2 B^2}{4 B_0}$$

$$\langle 1^{(0)} | 0^{(1)} \rangle = \frac{\langle 1^{(0)} | \hat{H}' | 0^{(0)} \rangle}{E_0^{(0)} - E_1^{(0)}} = \frac{-\hbar\gamma B}{2} \frac{1}{-\hbar\gamma B_0} = \frac{B}{2B_0}$$

$$\text{so } E_0 = -\frac{\hbar\gamma B_0}{2} - \frac{\hbar\gamma^2 B^2}{4 B_0} + \dots$$

$$|0\rangle = |+_z\rangle + \frac{B}{2B_0} |-_z\rangle + \dots$$

Exact: Recall that we can rotate a spinor into a new basis by

$$|\vec{e}_n \cdot \vec{S} : +\rangle = \begin{pmatrix} \cos \frac{\beta}{2} e^{-i\alpha/2} \\ \sin \frac{\beta}{2} e^{i\alpha/2} \end{pmatrix} \quad \text{where } \vec{n} = B_0 \vec{e}_z + B \vec{e}_x \Rightarrow \tan \beta = \frac{B}{B_0}$$

$$\vec{e}_n = \frac{\vec{n}}{|\vec{n}|} = \frac{B_0 \vec{e}_x + B \vec{e}_z}{\sqrt{B_0^2 + B^2}}$$

$$S_0 E_0 = \frac{-\hbar\gamma}{2} \sqrt{B_0^2 + B^2} = \frac{-\hbar\gamma B_0}{2} \left[1 + \frac{B^2}{2B_0^2} + \mathcal{O}(B^4) \right]$$

$$= \frac{-\hbar\gamma B_0}{2} - \frac{\hbar\gamma}{2} \frac{B^2}{B_0} + \mathcal{O}(B^4) \quad \checkmark$$

$$\text{and } |0\rangle = \cos \frac{\beta}{2} \begin{pmatrix} 1 \\ \tan \frac{\beta}{2} \end{pmatrix}$$

$$= \cos \frac{\beta}{2} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \left(\frac{B}{2B_0} \right) \right] = |0\rangle + \frac{B}{2B_0} |1^{(0)}\rangle + \dots$$

$\uparrow$
= 1 to order β

$\checkmark$

② Shankar 17.23

proton has charge $+e$.

$$\begin{aligned} 1) \text{ By Gauss' law, } \vec{E} &= \frac{+e}{r^2} \vec{e}_r \text{ for } r > R \\ &= \frac{+e}{r^2} \left(\frac{r^3}{R^3} \right) = \frac{+er}{R^3} \text{ for } r < R \end{aligned}$$

$$\begin{aligned} \text{so } \Phi(r) &= \int_r^\infty dr' E(r') = \frac{+e}{r} \text{ for } r > R \\ &= \frac{+e}{r} + \int_r^R dr' \left(\frac{+er'}{R^3} \right) = \frac{+3}{2} \frac{e}{R} - \frac{er^2}{2R^3} \text{ for } r < R \end{aligned}$$

Potential energy $V(r) = -e\Phi(r)$

$$= \begin{cases} -\frac{e^2}{r} & r > R \\ -\frac{3}{2} \frac{e^2}{R} + \frac{e^2 r^2}{2R^3} & r < R \end{cases}$$

$$2) \hat{H} = \hat{H}_0 + \hat{H}' \quad \text{let } \hat{H}_0 = \frac{-e^2}{r} \equiv V_0(r)$$

$$\text{Then } \hat{H}' = V(r) - V_0(r)$$

$$= 0 \quad r > R$$

$$= -\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \quad r < R$$

$$\text{Energy shift } E_0^{(1)} = \langle 100 | \hat{H}' | 100 \rangle$$

$$= \int_0^R dr r^2 \frac{4e^{-2r/a_0}}{a_0^3} \left(-\frac{3}{2} \frac{e^2}{R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \right)$$

Now take $e^{-2r/a_0} \approx 1$.

$$E_0^{(1)} \approx \frac{4}{a_0^3} \int_0^R dr \left(-\frac{3}{2} \frac{e^2 r^2}{R} + \frac{e^2 r^4}{2R^3} + e^2 r \right)$$

$$= \frac{4}{a_0^3} \left(-\frac{3e^2}{2R} \frac{R^3}{3} + \frac{e^2}{2R^3} \frac{R^5}{5} + e^2 \frac{R^2}{2} \right)$$

$$= \frac{2e^2 R^2}{5a_0^3}$$

③ Shankar 17.3.2

$$\hat{H}_0 = A \hat{S}_z^2 \quad \hat{H}' = B(\hat{S}_x^2 - \hat{S}_y^2)$$

$\hat{H}_0$ has 2 eigenvalues: 0 (nondegenerate: $|0\rangle$ in S_z basis)
 $A\hbar^2$ (2-fold degenerate: $|+\rangle$ and $|-\rangle$ in S_z basis)

Consider the degenerate subspace in the S_z basis: $|+\rangle, |-\rangle$.
 Diagonalize $\hat{H}'$ in this subspace:

$$\hat{H}' \Leftrightarrow \begin{pmatrix} H'_{++} & H'_{+-} \\ H'_{-+} & H'_{--} \end{pmatrix} \quad \text{where } H'_{+-} = \langle + | \hat{H}' | - \rangle \text{ etc.}$$

$$\text{In } \hat{S}_z \text{ basis, } \hat{H}' \Leftrightarrow B\hbar^2 \left[\begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}^2 - \begin{pmatrix} 0 & \frac{i}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} \\ 0 & \frac{i}{\sqrt{2}} & 0 \end{pmatrix}^2 \right]$$

$$= B\hbar^2 \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

so $\hat{H}' \Leftrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} B\hbar^2$ in sub-basis. This maximally

mixes the $S_z = \pm\hbar$ states; its eigenvectors are

$\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, which are the stable states under the (1st order) perturbation.

Exact states: in S_z basis, full Hamiltonian is just

$$\hat{H}_0 \leftrightarrow \hbar^2 \left[\begin{pmatrix} A & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & A \end{pmatrix} + \begin{pmatrix} 0 & 0 & B \\ 0 & 0 & 0 \\ B & 0 & 0 \end{pmatrix} \right] = \hbar^2 \begin{pmatrix} A & 0 & B \\ 0 & 0 & 0 \\ B & 0 & A \end{pmatrix}$$

Which has the same eigenvectors
as indicated by perturbation calculation: $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$, $\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$

④ Shankar 17.3.3

Spin-orbit correction is $\frac{1}{4} m c^2 \alpha^4 \frac{j(j+1) - l(l+1) - \frac{3}{4}}{n^3 l(l+\frac{1}{2})(l+1)}$

$$= \frac{1}{4} m c^2 \alpha^4 \frac{l \delta_{j, l+\frac{1}{2}} - (l+1) \delta_{j, l-\frac{1}{2}}}{n^3 l(l+\frac{1}{2})(l+1)}$$

Now consider perturbation $\hat{H} = -\hat{\mu}_z B$

$$\hat{\mu}_z = \frac{e}{2mc} \hat{L}_z + \frac{ge}{2mc} \hat{S}_z \quad \text{Take } g \approx 2.$$

$$\text{So } \hat{\mu}_z = \frac{e}{2mc} (\hat{L}_z + 2\hat{S}_z)$$

$$= \frac{e}{2mc} (\hat{J}_z + \hat{S}_z) \quad \text{is coupled}$$

$$\hat{H}' = -\frac{eB}{2mc} (\hat{J}_z + \hat{S}_z) \quad \text{but of course } \hat{J}_z, \hat{S}_z \text{ don't commute.}$$

First order shift = expectation value

$$\frac{-eB}{2mc} \langle \hat{J}_z + \hat{S}_z \rangle_{j=l\pm\frac{1}{2}, m_j}$$

$$= \frac{-eB}{2mc} \left[m_j \hbar + \langle \hat{S}_z \rangle \right] = \frac{-eB}{2mc} \left[j \hbar + \langle \hat{S}_z \rangle \right]$$

Need $|j=l\pm\frac{1}{2}, m\rangle$ in S_z basis:

$$|j=l\pm\frac{1}{2}, m\rangle = C_+ |m_l=m_j-\frac{1}{2}, m_s=\frac{1}{2}\rangle + C_- |m_l=m_j+\frac{1}{2}, m_s=-\frac{1}{2}\rangle$$

$$C_{\pm} \text{ Clebsch-Gordan coeffs are } C_+ = \pm \sqrt{\frac{l\pm m + \frac{1}{2}}{2l+1}} \quad C_- = \sqrt{\frac{l\mp m + \frac{1}{2}}{2l+1}}$$

$$\langle \hat{S}_z \rangle = \frac{\hbar}{2} |C_+|^2 - \frac{\hbar}{2} |C_-|^2 = \frac{\hbar}{2(2l+1)} \left(l\pm m + \frac{1}{2} - l\pm m - \frac{1}{2} \right) = \frac{\pm m \hbar}{2l+1} = \frac{\pm j \hbar}{2l+1}$$

$$\Rightarrow E \text{ shift} = \frac{-eB}{2mc} \left[1 \pm \frac{1}{2l+1} \right] j \hbar \quad \text{QED.}$$

5

$$\hat{H}_0 = \frac{\hat{p}_x^2}{2m} + \frac{\hat{p}_y^2}{2m} + \frac{1}{2}m\omega_0(\hat{x}^2 + \hat{y}^2)$$

Define quantum numbers n_x, n_y and operators

$\hat{a}_x, \hat{a}_x^\dagger, \hat{a}_y, \hat{a}_y^\dagger$ as usual.

The eigenvalue $3\hbar\omega_0$ is 3-fold degenerate.

In the $\hat{H}_0$ basis, degenerate subspace is spanned by states $|n_x=1, n_y=1\rangle, |n_x=0, n_y=2\rangle, |n_x=2, n_y=0\rangle \equiv |11\rangle, |02\rangle, |20\rangle$

$$\begin{aligned}\hat{H}' &= 2\lambda\hat{x}\hat{y} = \frac{\hbar\lambda}{m\omega_0}(\hat{a}_x + \hat{a}_x^\dagger)(\hat{a}_y + \hat{a}_y^\dagger) \\ &= \frac{\hbar\lambda}{m\omega_0}(\hat{a}_x\hat{a}_y + \hat{a}_x^\dagger\hat{a}_y + \hat{a}_y^\dagger\hat{a}_x + \hat{a}_x^\dagger\hat{a}_y^\dagger)\end{aligned}$$

Now let a generic vector $|\psi\rangle = \psi_1|11\rangle + \psi_2|02\rangle + \psi_3|20\rangle$ in the degenerate subspace. Need to find $|\psi\rangle$ that are eigenvectors of $\hat{H}'$ (in the subspace).

$$\begin{aligned}\hat{H}'|\psi\rangle &= \psi_1\hat{H}'|11\rangle + \psi_2\hat{H}'|02\rangle + \psi_3\hat{H}'|20\rangle \\ &= \frac{\hbar\lambda}{m\omega_0} \left\{ \psi_1 \left[|00\rangle + \sqrt{2}|02\rangle + \sqrt{2}|20\rangle + 2|22\rangle \right] + \psi_2 \left[0 + \sqrt{2}|11\rangle + \sqrt{3}|13\rangle \right] \right. \\ &\quad \left. + \psi_3 \left[0 + \sqrt{2}|11\rangle + \sqrt{3}|31\rangle \right] \right\}\end{aligned}$$

Ignore ~~terms~~ terms outside subspace: $\hat{H}'_{\text{sub}} \leftrightarrow \begin{pmatrix} 0 & \sqrt{2} & \sqrt{2} \\ \sqrt{2} & 0 & 0 \\ \sqrt{2} & 0 & 0 \end{pmatrix} \frac{\hbar\lambda}{m\omega_0}$

which has eigenvalues $[-2, 0, +2] \frac{\hbar\lambda}{m\omega_0}$.

⑤ cont'd

Ground state shift: $\langle 00 | \hat{H}' | 00 \rangle$

$$= \frac{\hbar \lambda}{m \omega_0} \langle 00 | \hat{a}_x^\dagger \hat{a}_y^\dagger | 00 \rangle \quad (\text{other three terms annihilate } |00\rangle)$$

$$= \frac{\hbar \lambda}{m \omega_0} \langle 00 | 11 \rangle = 0$$

So ground state is unchanged to first order.

⑥ Exact $\hat{H}$:

$$\hat{H} = \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2) + \frac{1}{2} m \omega^2 (\hat{x}^2 + \hat{y}^2) + 2\lambda \hat{x} \hat{y}$$

$$= \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2) + \frac{1}{2} m \omega^2 \left[\left(\frac{\hat{x}^2}{2} + \frac{\hat{y}^2}{2} + \hat{x} \hat{y} \right) + \left(\frac{\hat{x}^2}{2} + \frac{\hat{y}^2}{2} - \hat{x} \hat{y} \right) + \frac{4\lambda \hat{x} \hat{y}}{m \omega^2} \right]$$

$$= \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2) + \frac{1}{2} m \omega^2 \left[\left(\frac{\hat{x}^2}{2} + \frac{\hat{y}^2}{2} + \hat{x} \hat{y} + \frac{\lambda}{m \omega^2} \hat{x}^2 + \frac{\lambda}{m \omega^2} \hat{y}^2 + \frac{2\lambda \hat{x} \hat{y}}{m \omega^2} \right) + \left(\frac{\hat{x}^2}{2} + \frac{\hat{y}^2}{2} - \hat{x} \hat{y} - \frac{\lambda}{m \omega^2} \hat{x}^2 - \frac{\lambda}{m \omega^2} \hat{y}^2 + \frac{2\lambda \hat{x} \hat{y}}{m \omega^2} \right) \right]$$

$$= \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2) + \frac{1}{2} m \omega^2 \left[\left(1 + \frac{\lambda}{m \omega^2} \right) \left(\frac{\hat{x} + \hat{y}}{\sqrt{2}} \right)^2 + \left(1 - \frac{\lambda}{m \omega^2} \right) \left(\frac{\hat{x} - \hat{y}}{\sqrt{2}} \right)^2 \right]$$

$$\text{Let } u \equiv \frac{x+y}{\sqrt{2}} \quad v \equiv \frac{x-y}{\sqrt{2}}$$

$$= \frac{1}{2m} (\hat{p}_u^2 + \hat{p}_v^2) + \frac{1}{2} m \omega^2 \left[\left(1 + \frac{\lambda}{m \omega^2} \right) \hat{u}^2 + \left(1 - \frac{\lambda}{m \omega^2} \right) \hat{v}^2 \right]$$

$$\Rightarrow \text{Define } \omega_u^2 = \omega^2 \left(1 + \frac{\lambda}{m \omega^2} \right) \quad \omega_v^2 = \omega^2 \left(1 - \frac{\lambda}{m \omega^2} \right)$$

To first order in λ , $\omega_u = \omega + \frac{\lambda}{2m\omega}$, $\omega_v = \omega - \frac{\lambda}{2m\omega}$

So the eigenvalues of $\hat{H}$ are $\left[\omega_u(n_u + \frac{1}{2}) + \omega_v(n_v + \frac{1}{2})\right]\hbar$

$$= \hbar \left[(n_u + n_v + 1)\omega + \frac{\lambda}{2m\omega} (n_u - n_v) \right]$$

So the shift for the ground state ($n_u = n_v = 0$) is zero,

for the 1st excited states:

$$n_u = n_v = 1: \text{ zero}$$

$$n_u = 2, n_v = 0: \frac{\lambda\hbar}{m\omega}$$

$$n_u = 0, n_v = 2: -\frac{\lambda\hbar}{m\omega}$$

As expected from pert. theory calculation.